

Final Report

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1 Introduction

Epilepsy diagnosis requires specialists to manually review hours of continuous EEG, making the process slow and limiting assessment outside clinical hours. This project classifies 12-second EEG segments as seizure or non-seizure and uses Grad-CAM to highlight influential regions. Deep learning can detect subtle seizure activity by using a CNN to learn spatial patterns and an LSTM to track their evolution over time. The project evaluates how performance scales with the number of training patients under a strict patient-level split, addressing evidence that many CHB-MIT studies do not rigorously separate training and evaluation patients.

2 Illustration

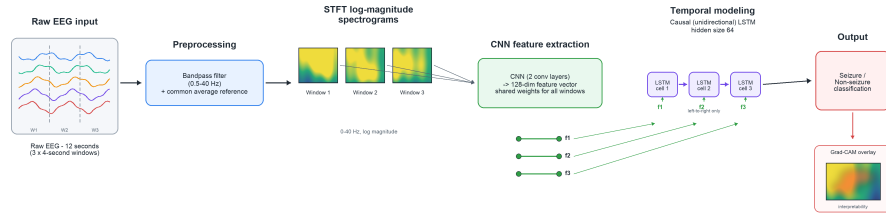


Figure 1: Architecture of the lightweight CNN-LSTM seizure detection pipeline.

3 Background and Related Work

Prior work has combined CNNs and LSTMs for seizure detection on this dataset using bidirectional LSTMs which require future EEG windows and are there-

fore not deployable for real-time monitoring [5]. CNN approaches with Grad-CAM interpretability have also been applied to this dataset [1]. Other work has evaluated window sizes from 4 to 12 seconds on this dataset, including a ResNet-LSTM architecture. [2] Some prior work reports almost perfect accuracy on this dataset. Given that seizure windows represent only 1–2% of the data, such results should be interpreted cautiously and motivate the use of sensitivity, specificity, and F1 rather than accuracy as primary metrics [4]. A recent study documents that many existing studies using this dataset do not rigorously separate patients across training and evaluation risking inflated performance estimates. This directly motivates the strict patient-level split and the scaling experiment central to this project [3].

4 Data Processing

This project uses the CHB-MIT Scalp EEG Database, which contains pediatric EEG recordings sampled at 256 Hz using a 23-channel bipolar montage. After standardizing inconsistent channel counts, 22 channels were retained. Patients 1–18 were used for training, 19–21 for validation, and 22–24 for final testing. Recordings were filtered to 0.5–40 Hz, common-average re-referenced to reduce noise, and divided into 4-second windows labeled as seizures when overlapping a documented seizure by at least one second. Each window was converted into a log-magnitude spectrogram using the short-time Fourier transform, retaining frequencies up to 40 Hz. Three consecutive windows were combined into overlapping 12-second sequences, labeled as seizures if any window contained seizure activity. Only 1.8% of training sequences were seizure-labeled, demonstrating severe class imbalance.

5 Architecture

The primary model is a lightweight CNN-LSTM with 120,289 parameters. Two convolutional layers with batch normalization, ReLU, and max pooling, plus adaptive average pooling and a linear layer, produce a 128-dimensional feature vector for each spectrogram. A causal, unidirectional LSTM with a hidden size of 64 processes the three window features using only past and present information, and a linear classifier outputs one logit per sequence. The proposed pretrained ResNet-18 with the same LSTM was implemented and its forward pass verified, but compute constraints prevented full-scale training.

6 Baseline Model

The baseline is a hand-coded amplitude-threshold detector with no learned parameters. A single global amplitude threshold (mean amplitude 33.11, standard deviation 51.01) was calculated using only the training patients. This fixed threshold was then applied to the validation set, mirroring the cross-patient

evaluation of the primary model. A window was flagged as seizure if at least two channels exceeded the mean plus 2.5 standard deviations for at least two continuous seconds. This baseline achieved a validation sensitivity of 0.189, specificity of 0.999, precision of 0.712, and F1 of 0.298.

7 Quantitative Results

The primary model, trained on 6 patients, achieved a validation F1 of 0.480, sensitivity of 0.403, specificity of 0.994, and precision of 0.595, exceeding the baseline’s F1 of 0.298, sensitivity of 0.189, specificity of 0.999, and precision of 0.712. On the held-out test set, the same model achieved F1 0.284, sensitivity 0.371, specificity 0.979, and precision 0.230. The central experiment of this project measures validation F1 as a function of training patient count: 0.266 at 3 patients, 0.480 at 6, and 0.420 at 12.

8 Qualitative Results

Grad-CAM was applied to the final convolutional layer of the trained model to visualize which time-frequency regions most influenced each prediction. Figure 2 shows a correctly detected seizure sequence (predicted probability 0.9993) from a held-out test patient, illustrating the time-frequency pattern the model relies on for confident, correct detections. A confident false positive (predicted probability 1.0000) and a false negative from within an annotated seizure event were also examined; together with the true positive shown, these cases illustrate that the model’s failure modes, occasionally mistaking non-seizure activity for seizure and missing part of a seizure’s duration, are what limit its precision and sensitivity on unseen patients.

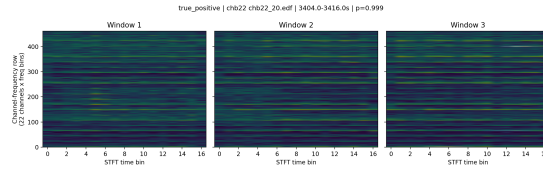


Figure 2: Grad-CAM visualization of a correctly detected seizure sequence highlighting the time-frequency regions the model attended to.

9 Evaluate Model on New Data

The trained model was evaluated once on patients 22–24, who were excluded from all preprocessing, training, and model-selection decisions. It achieved a sensitivity of 0.371, specificity of 0.979, precision of 0.230, and F1 score of 0.284.

Because overlapping windows were used, the 353 seizure-labeled sequences represent only 26 independent seizure events. All 26 events had at least one correctly flagged sequence, but the average within-event detection rate was 34.7%, meaning the model detected each seizure without consistently tracking its full duration. Results were consistent across patients, with individual sensitivities of 0.36–0.38 and specificities of 0.974–0.985, confirming that no single patient drove the pooled performance.

10 Discussion

The F1 drop from validation (0.480) to the held-out test set (0.284) shows that performance on training-distribution patients overestimates generalization to unseen patients, reinforcing the need for patient-level evaluation. Validation F1 improved from 3 to 6 patients but declined at 12 under the fixed four-epoch budget, suggesting the lightweight model could not fully learn the added patient variability. Event-level results showed that the model detected every seizure but did not consistently track its full duration. The proposed ResNet-18–LSTM was also implemented and its forward pass verified; a partial six-patient run reduced training loss but could not be completed after free-tier Colab GPU access became unavailable.

11 Ethical Considerations

Because this dataset contains only 24 pediatric patients, the model’s generalization across ages, demographics, and seizure causes remains unverified and requires direct validation in broader pediatric populations. Its 0.371 sensitivity on unseen patients is insufficient for standalone clinical decision-making, as missed seizures pose serious safety risks. Any real-world use should therefore be limited to an assistive role under clinician oversight.

12 Project Difficulty/Quality

This project presented challenges beyond a standard course assignment: seizure prevalence was below 2%, channel inconsistencies required programmatic standardization, and infrastructure instability required a transparent mid-project architecture change. The patient-scaling experiment and independent test set enabled rigorous evaluation without patient-level leakage, a known limitation in prior CHB-MIT studies. Given substantial variation in EEG patterns across patients, the modest held-out performance reflects the unresolved challenge of cross-patient generalization rather than an implementation weakness.

References

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